

L2 - 11.4. OpenCTI

Task 1 - Introduction

Threat intelligence is about more than just collecting IPs and hashes, as it requires understanding the who, how, and why behind an attack. OpenCTI provides a framework for mapping these connections, turning reports into a comprehensive understanding of the threat landscape. This room introduces you to the platform's core features and shows you how to navigate and analyze threat intelligence.

Task 2 - OpenCTI Overview

Managing Cyber Threat Intelligence (CTI) is a constant challenge, as organizations struggle to ingest, analyze, and present massive volumes of data effectively. It allows teams to store, analyze, and visualize complex threat landscapes, tracking the relationships between campaigns, malware, and IOCs back to their primary source. To build a unified view, the platform standardizes data using MITRE ATT&CK and seamlessly integrates with tools such as MISP and TheHive.

Operational Use Cases: Who Uses OpenCTI?

Threat intel covers everything from low-level artifacts to high-level organizational risk. Instead of letting that data sit disconnected, OpenCTI pulls it all into a unified knowledge graph. This gives different security teams a shared, readable view for hunting and tracking adversarial campaigns.

- **Government and national security:** To automate the sharing of IOCs, link incidents, prioritize threats, and coordinate responses
- **Enterprise and private sector:** To build internal threat intelligence, automate detection and hunting, and reduce response times
- **Cyber security service providers:** To centralize knowledge bases, deliver threat intelligence services to clients, and feed SIEM/EDR systems
- **Blue teams and incident response:** For threat hunting, correlating observables with campaigns and TTPs, incident management, and detection

OpenCTI and MISP

A range of open-source threat intelligence platforms has been made available to the cyber security community. Users who have interacted with MISP (Malware Information Sharing Platform) in the past might wonder how these platforms differ in their offerings. Rather than viewing them as competing tools, modern security teams typically treat them as complementary parts of a pipeline. For example, threat data is frequently ingested from MISP into OpenCTI via a one-way connector to build out a more comprehensive threat profile. Let's take a closer look at their offerings below:

Feature	MISP (Event-Focused)	OpenCTI (Knowledge-Focused)
Core question	What indicators should I block and share?	Who is attacking us, why, what tools are they using, and how does it map to our organization?
Data focus	Technical, flat lists of indicators of compromise (IOCs)	Strategic context connecting threats, campaigns, vulnerabilities and TTPs
Core structure	Lightweight, event-driven indicator correlation	Interconnected knowledge graph built on the STIX standard

Task 3 - The Intelligence Pipeline

Before getting hands-on with OpenCTI, let's cover how data is ingested into the platform and where the data comes from. To protect an organization, security analysts cannot rely solely on a flat list of bad IP addresses. They need to understand the who, how, and why behind cyber attacks. OpenCTI achieves this by shifting from traditional static lists to a dynamic knowledge graph. But how does this data get into the platform, and how does it make sense of it all?

Intelligence Sources

OpenCTI doesn't come with data preinstalled. It is a collaborative platform designed to ingest data from a range of intelligence feeds, research groups, and cyber teams worldwide.

- **Structured frameworks and repositories:** Uses the MITRE ATT&CK framework and the National Vulnerability Database (NVD) to standardize behavior and exploits

- **Threat intelligence platforms:** Open-source and community sharing platforms like MISP and AlienVault OTX feed indicators directly into the platform
- **Internal threat teams:** Your organization's private security researchers, incident response teams, and SOC analysts can manually upload custom incident files
- **Commercial threat intelligence feeds:** Premium subscription-based intelligence provided by security firms like CrowdStrike feeds data directly into OpenCTI
- **Infrastructure and OSINT trackers:** Open-source intelligence projects like Shodan and VirusTotal are monitored by platform connectors

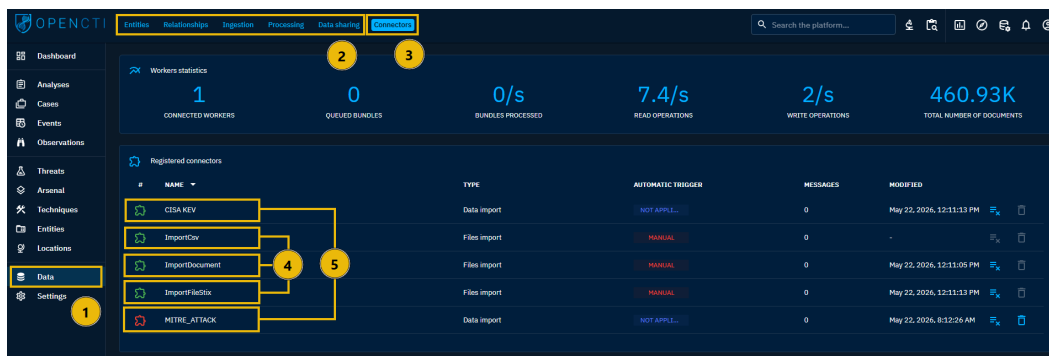
Getting the Data

To handle the various data sources, OpenCTI uses Connectors(opens in new tab) to ingest and enrich data. Connectors work in the background to fetch and normalize incoming data into a standardized layout. There are multiple connectors available, but analysts will typically use two ingestion types:

- **External input connectors:** Automatically pull threat feeds and community data into the platform
- **Internal import connectors:** Parse manually uploaded local documents like PDF reports and indicator CSVs

We can visualize this in our own OpenCTI instance by opening the **Data** tab in the sidebar menu:

1. Click **Data** in the left sidebar menu
2. Explore the available data types and ingestion frameworks
3. Select the **Connectors** tab from the sub-menu
4. Review the internal import connectors handling manual local file uploads
5. Review the listed external input connectors handling incoming live streams



The Data Language

Once the data passes through a connector, OpenCTI needs a unified way(opens in new tab) to map it. It achieves this by structuring incoming data using the industry-standard STIX 2.1 (Structured Threat Information Expression). You can think of STIX as a universal language that enables different security tools, platforms, and teams to exchange intelligence without translation issues. Let's take a look at the STIX naming objects below to get a grasp of STIX relationships and the types of graphs and language we will encounter in later tasks:

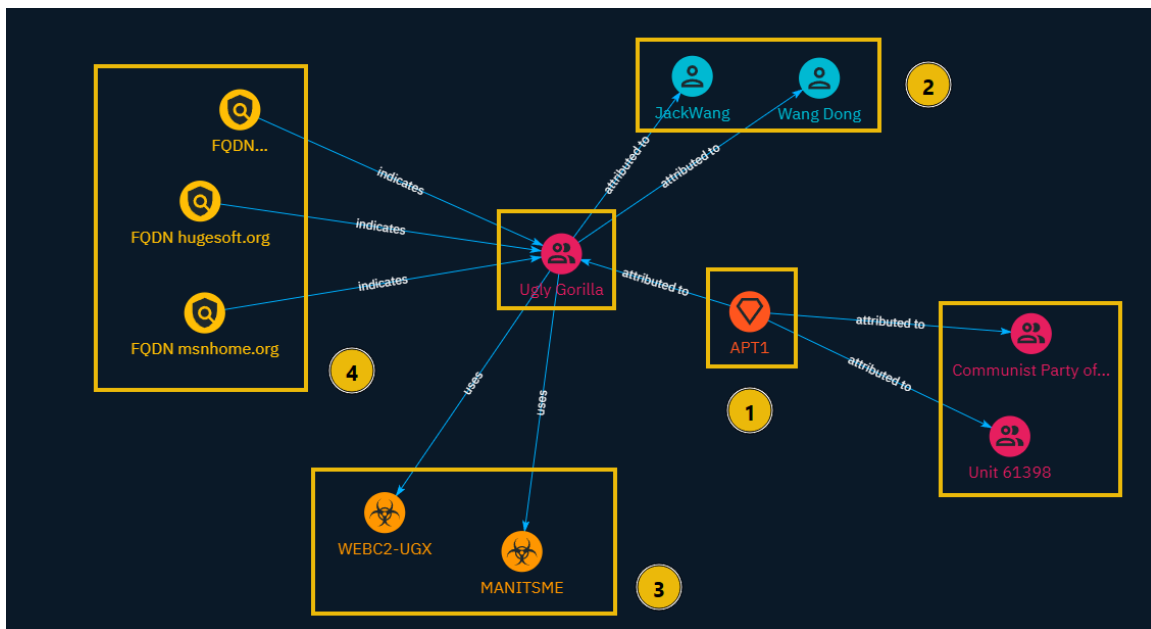
- **STIX Domain Object (SDO):** Represent threat objects such as attack patterns, threat actors, or a type of malware
- **STIX Cyber Observable (SCO):** Technical indicators, such as an IP address, a domain name, or a file hash
- **STIX Relationship Objects (SRO):** They map out the precise relationship between different SDOs and SCOs

Example

To visualize how these pieces lock together, let's look at the diagram below. We will start with a central threat and branch out from there.

1. Our core target is APT1. The graph shows relationships by directly attributing them to three other known threat actor aliases
2. We can pivot from organizational groups to individuals. Here, specific actors are directly attributed to the Ugly Gorilla cell
3. Next, the graph shows the specific malware families our target has been observed using in campaigns

4. Finally, further entity relationships are established by mapping the exact domain names that indicate Ugly Gorilla's presence

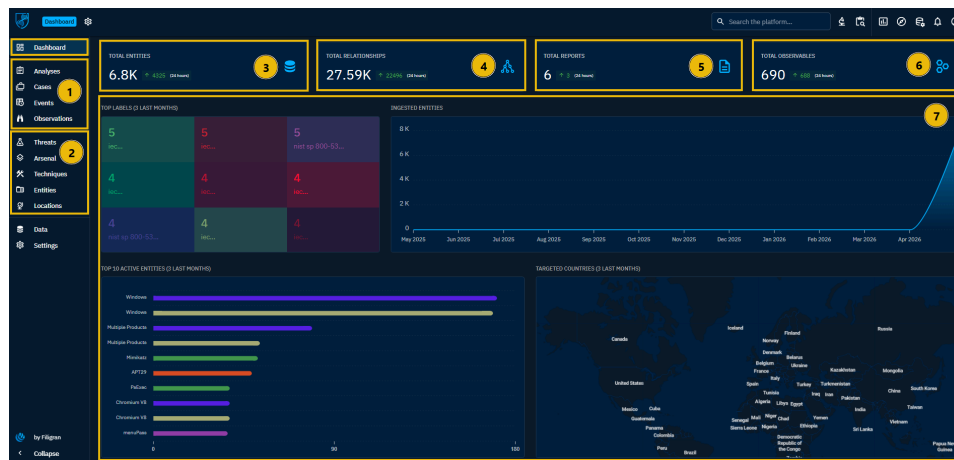


Task 4 - Navigating OpenCTI

Now that we have a solid understanding of what OpenCTI is and how data is ingested into the platform, let's get hands-on and begin exploring the user interface. When you first log into the web panel, OpenCTI presents an overview of all available intelligence data stored in your instance. This landing page displays high-level statistics, timelines, and geolocation data. It is worth noting that this dashboard is fully customizable and can be tailored to your organization's specific needs.

1. **Hot knowledge:** Entities and relationships that require analysis
2. **Cold knowledge:** The entities and relationships used for hot knowledge
3. **Entities:** Threat actors, malware families, targeted sectors
4. **Relationships:** Connections that link the data points together
5. **Reports:** Formalized intelligence documents or briefs
6. **Observables:** Raw artifacts such as IP addresses and file hashes

7. Overview: The top labels, active entities, and targeted countries



Exploring Hot Knowledge: This section includes items that require analysis and contains information added by users or the organization managing the OpenCTI instance.

Analyses: Review the Analyses section in MISP to explore threat reports, malware analysis, and select the APT1 report.

1. Click **Analyses**
2. See all types of data ready for analysis
3. Select the **APT1** threat report

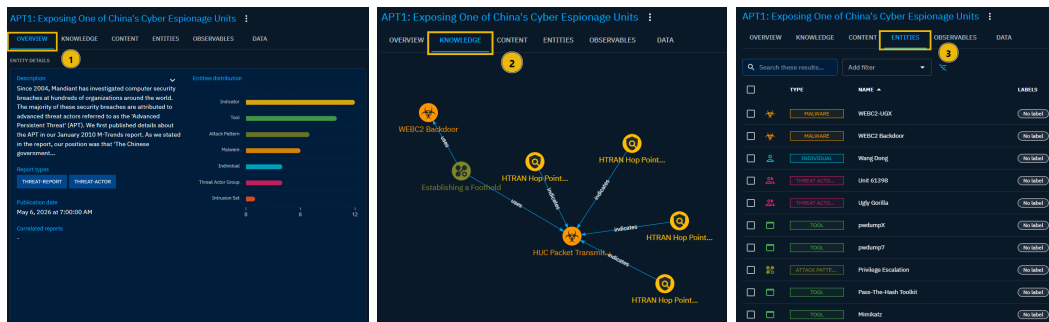
The image shows the OpenCTI Reports table. The 'Reports' tab is selected in the top navigation bar. The table lists various threat reports with columns for Title, Type, Author, Creators, Labels, Date, Status, and Marking. The 'APT1: Exposing One of China's Cyber Espionag...' report is highlighted with a yellow box.

	TITLE	TYPE	AUTHOR	CREATORS	LABELS	DATE	STATUS	MARKING
☐	Cloudflare Threat Report 2026	threat-re...	-	admin	No label	May 20, 2026	DISABLED	NONE
☐	Elastic Threat Report 2025	threat-re...	-	admin	No label	May 20, 2026	DISABLED	NONE
☐	M Trends Mandiant 2026	threat-re...	-	admin	No label	May 20, 2026	DISABLED	NONE
☐	APT1: Exposing One of China's Cyber Espionag...	threat-re...	-	admin	No label	May 6, 2026	DISABLED	NONE
☐	Poison Ivy: Assessing Damage and Extracting L...	threat-re...	FireEye, Inc.	admin	No label	May 6, 2026	DISABLED	NONE
☐	APT37	threat-re...	-	admin	No label	May 4, 2022	DISABLED	NONE

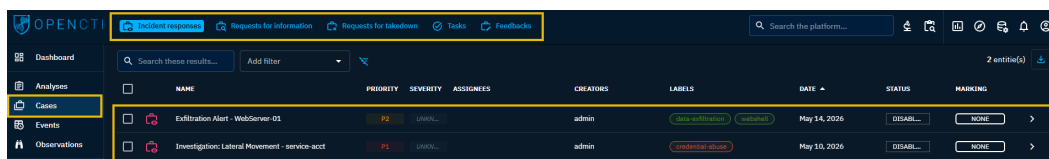
Once you've selected a report or other item for analysis, you'll have a number of options for further investigation. Let's take a look at a few interesting sections below:

1. **Overview:** Provides a high-level summary of the object or report

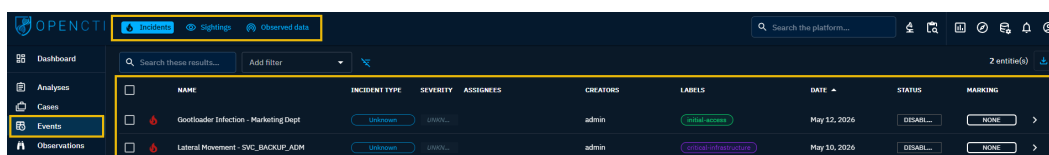
2. **Knowledge:** Serves as the central core of the analysis, visualizing linked intelligence data in a graph format
3. **Entities:** Displays a structured breakdown of all the specific entities, including tools, groups, malware, and tactics



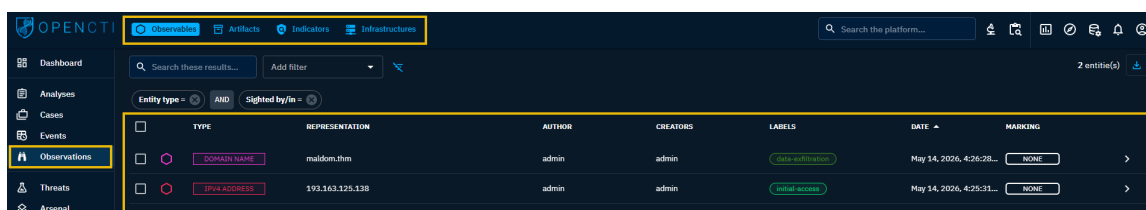
Cases: Use the Cases section as a built-in case management tool to organize incident response, track investigations, and consolidate evidence.



Events: Use the Events section as a centralized log for live security telemetry and alerts to track indicators against threat intelligence profiles.



Observations: Explore the Observations section in OpenCTI to manage technical telemetry, observables like IP addresses, malware artifacts, and detection rules.

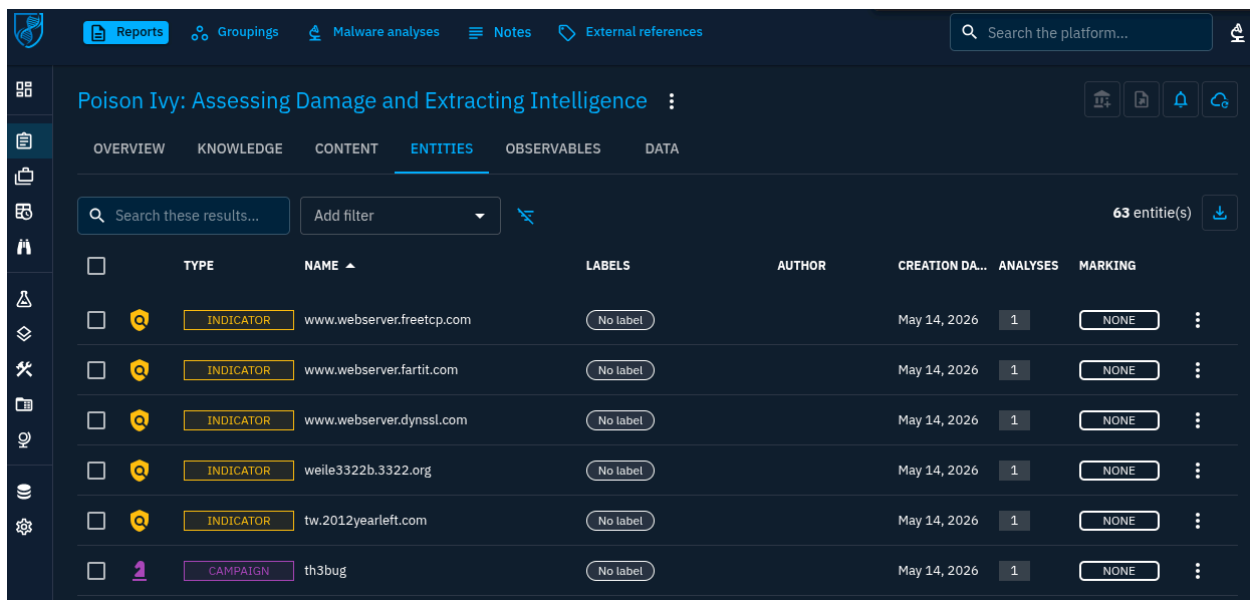


Answer the questions below

Check out the Poison Ivy threat report within the Analyses tab. Which campaign relied on the use of the Strategic Web Compromise attack pattern?

th3bug

Go to the **analysis section**, select **Poison Ivy**, and open the **entities** tab—the campaign will be listed there.



The screenshot shows the 'Poison Ivy: Assessing Damage and Extracting Intelligence' report in the 'ENTITIES' tab. The table lists 63 entities, with the following visible rows:

	TYPE	NAME	LABELS	AUTHOR	CREATION DA...	ANALYSES	MARKING
☐	INDICATOR	www.webserver.freetcp.com	No label		May 14, 2026	1	NONE
☐	INDICATOR	www.webserver.fartit.com	No label		May 14, 2026	1	NONE
☐	INDICATOR	www.webserver.dynssl.com	No label		May 14, 2026	1	NONE
☐	INDICATOR	weille3322b.3322.org	No label		May 14, 2026	1	NONE
☐	INDICATOR	tw.2012yearleft.com	No label		May 14, 2026	1	NONE
☐	CAMPAIGN	th3bug	No label		May 14, 2026	1	NONE

When viewing the threat intelligence report for APT37, how many distinct malware entities are listed?

8

Navigate to the **analysis section**, select **APT37**, open the **entities** tab, and you will find 8 malwares listed there.

TYPE	NAME	LABELS	AUTHOR	CREATION DA...	ANALYSES	MARKING
MALWARE	Wiper	No label	The MITRE Corp...	May 14, 2026	2	Copyrig...
MALWARE	WINERACK	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...
MALWARE	SLOWDRIFT	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...
MALWARE	SHUTTERSPEED	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...
MALWARE	POORAIM	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...
MALWARE	HAPPYWORK	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...
MALWARE	DOGCALL	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...
MALWARE	CORALDECK	No label	The MITRE Corp...	May 14, 2026	1	Copyrig...

Which directory path was targeted in the Exfiltration Alert case?

/var/www/html/uploads/

Navigate to the **case section**, select **Exfiltration**, and read the description.

Priority

P2

Severity

UNKNO...

Incident response type

-

Description

Outbound connection spike detected from WebServer-01 to a high-risk external IP. Evidence of web shell persistence found in </var/www/html/uploads/>.

Correlated cases

-

Marking

NONE

Author

ADMIN

Reliability (of author)

Unknown

Confidence level

1 - Confirmed by other sources

Distribution of opinions

NO

Labels

data-exfiltrati

Creation date (in

May 14, 2026

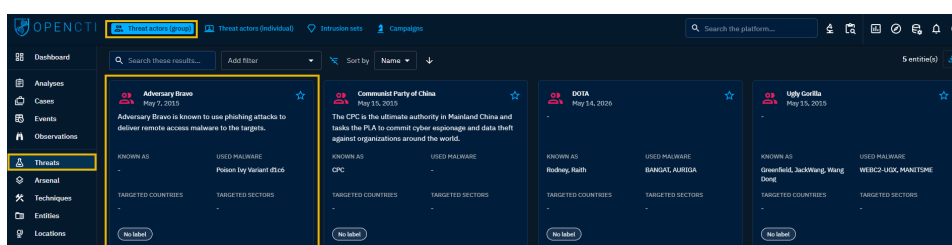
Task 5 - Exploring Cold Knowledge

Now that we have explored the first section of the OpenCTI interface, it is time to discover where that data originates. The second major navigational component, known as Cold Knowledge, serves as the platform's structural encyclopedia. While your organization's telemetry changes by the minute, this section tracks the permanent background details of the threat landscape.

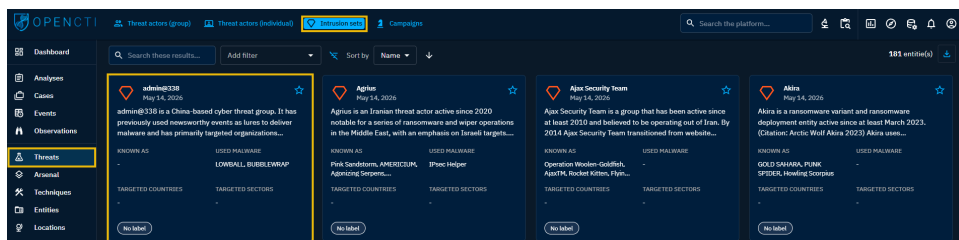
Cold Knowledge Breakdown

Threats: The [Threats \(opens in new tab\)](#) section gives you an overview of various threat actors, both groups and individuals, intrusion sets, and campaigns carried out by the threat actors.

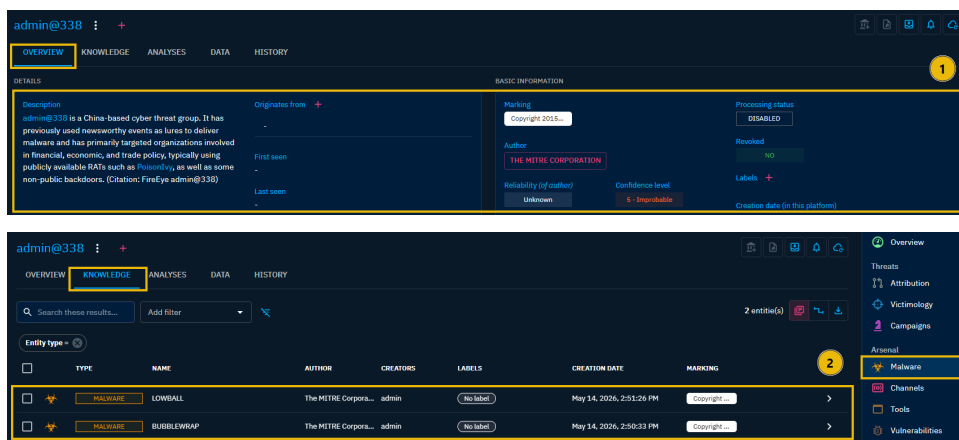
- **Threat actors (groups):** Actual structured malicious organizations, nation-state intelligence agencies, or other cybercriminal groups
- **Threat actors (individuals):** Single, specific human beings known or suspected to be executing malicious cyber activity
- **Intrusion sets:** A tracked baseline of recurring technical behaviors, infrastructure patterns, malware toolkits, and adversarial habits that target similar victims. These sets can be linked to specific threat actors
- **Campaigns:** Attacks occurring over a specific period of time or targeting a specific sector or organization



Clicking on any entity within this section will allow you to access all knowledge available for the particular entity. Let's try it out by navigating to **Intrusion sets** and selecting **admin@338**.

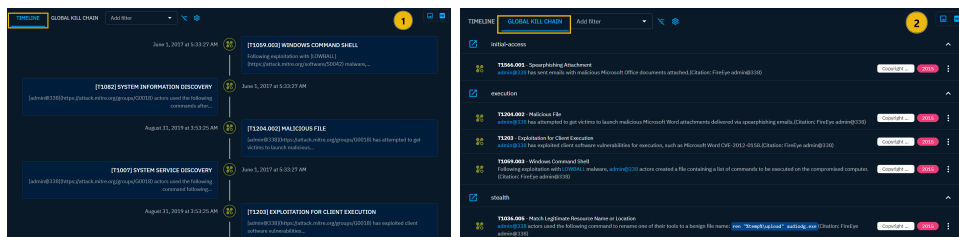


1. First, we will see the **Overview** section, similar to what we saw in the Analyses section previously
2. Selecting **Knowledge** allows you to see all data associated with the intrusion set, including campaigns, indicators, and attack patterns

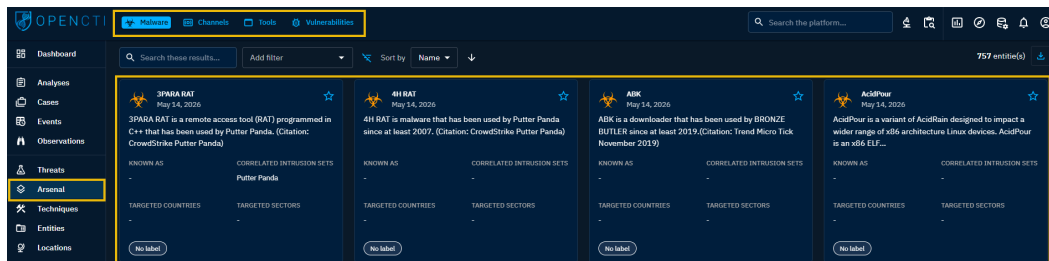


Scrolling down in the Knowledge section reveals two further sections worth noting:

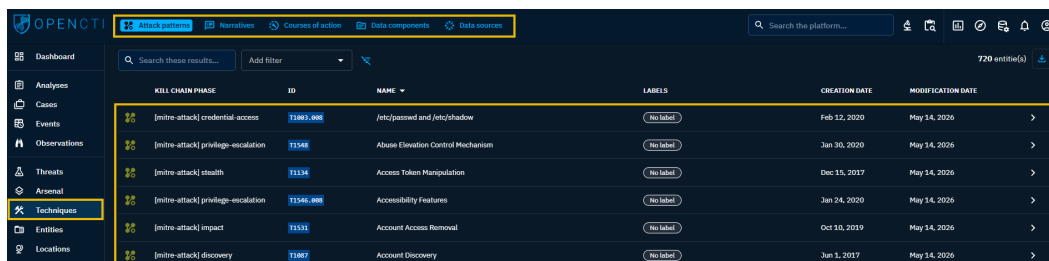
1. **Timeline**: Provides a chronological view of all related events, relationships, and activities linked to the entity
2. **Global Kill Chain**: Visually shows where the threat actor fits into the different stages of an attack



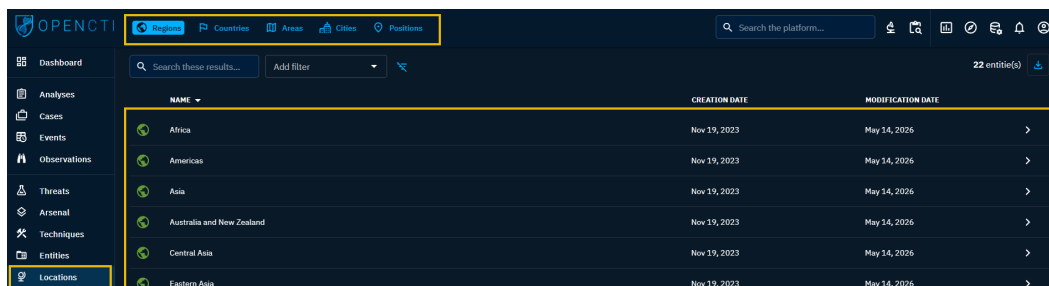
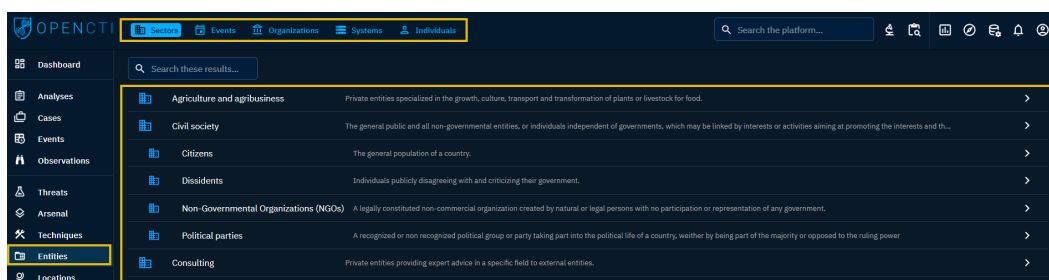
Arsenal: Explore the Arsenal section in OpenCTI to access technical documentation on active malware families, malicious tools, and mapped CVE exploits.



Techniques: Explore the Techniques section in OpenCTI to review MITRE ATT&CK attack patterns, adversary behaviors, and defensive mitigation strategies.



Entities and Locations: Use the Entities and Locations sections in OpenCTI to map targeted organizations and geographic tracking data to understand intrusion contexts.



Answer the questions below

What command has Andariel used during the Discovery phase to display TCP connections on target servers?

netstat -naop tcp

Select **Andriel** in the **threat section**, navigate to the **knowledge** tab, and view the **1 execution** listed there.

The screenshot shows the MITRE ATT&CK interface for the threat actor 'Andariel'. The 'OVERVIEW' tab is selected, displaying details about the threat actor. The 'DESCRIPTION' section states: 'Andariel is a North Korean state-sponsored threat group that has been active since at least 2009. Andariel has primarily focused its operations--which have included destructive attacks--against South Korean government agencies, military organizations, and a variety of domestic companies; they have also conducted cyber financial operations against ATMs, banks, and cryptocurrency exchanges. Andariel's notable activity includes Operation Black Mine, Operation GoldenAxe, and...'. The 'BASIC INFORMATION' section includes: 'Marking: Copyright 2015...', 'Author: THE MITRE CORPORATION', 'Reliability (of author): Unknown', 'Confidence level: 5 - Improbable', 'Processing status: DISABLED', 'Revoked: NO', 'Labels: +', 'Creation date (in this platform): May 14, 2026 at 10:05:31 AM', 'Creators: ADMIN', 'Standard STIX ID: intrusion-set--4bdf96b-8fb8-5c4e-82dc...', 'Creation date: September 29, 2021 at 6:10:19 PM', and 'Modification date: May 14, 2026 at 10:05:31 AM'.

The screenshot shows the 'KNOWLEDGE' tab for the threat actor 'Andariel'. It displays a timeline of activities. The first activity is '[T1203] EXPLOITATION FOR CLIENT EXECUTION' on September 29, 2021 at 6:41:18 PM, with the description: '[Andariel](https://attack.mitre.org/groups/G0138) has exploited numerous ActiveX vulnerabilities,...'. The second activity is '[T1049] SYSTEM NETWORK CONNECTIONS DISCOVERY' on September 29, 2021 at 6:41:18 PM, with the description: '[Andariel](https://attack.mitre.org/groups/G0138) has used the netstat -naop tcp ` command to...'. The third activity is 'RIFDOOR' on September 29, 2021 at 6:41:18 PM, with the citation: '(Citation: AhnLab Andariel Subgroup of Lazarus June 2018)'.

Using the Arsenal section, which threat group is noted as using the DCRAT tool?

APT-C-36

Search for **DCRAT** in the arsenal section.

The screenshot shows a web interface for a security tool named DCRAT. The top navigation bar includes links for Malware, Channels, Tools (which is highlighted), and Vulnerabilities. On the left, there is a sidebar with various icons. The main content area is titled 'DCRAT' and has tabs for OVERVIEW, KNOWLEDGE, ANALYSES, DATA, and HISTORY. The OVERVIEW tab is selected, showing details about the tool. The details section includes a description, tool types, kill chain phases, and tool version. Below this, there is a section for 'LATEST CREATED RELATIONSHIPS' which lists several relationships between the tool and other entities. The bottom row of this list is highlighted with a red circle, showing a relationship between DCRAT and APT-C-36.

DCRAT : +

OVERVIEW KNOWLEDGE ANALYSES DATA HISTORY

DETAILS

Description
DCRAT is a variant of the open-source AsyncRAT developed in C# with additional capabilities such as patching Microsoft's Antimalware Scan Interface (AMSI). (Citation: Zscaler BlindEagle DEC 2025)

Tool types
-

Kill chain phases
-

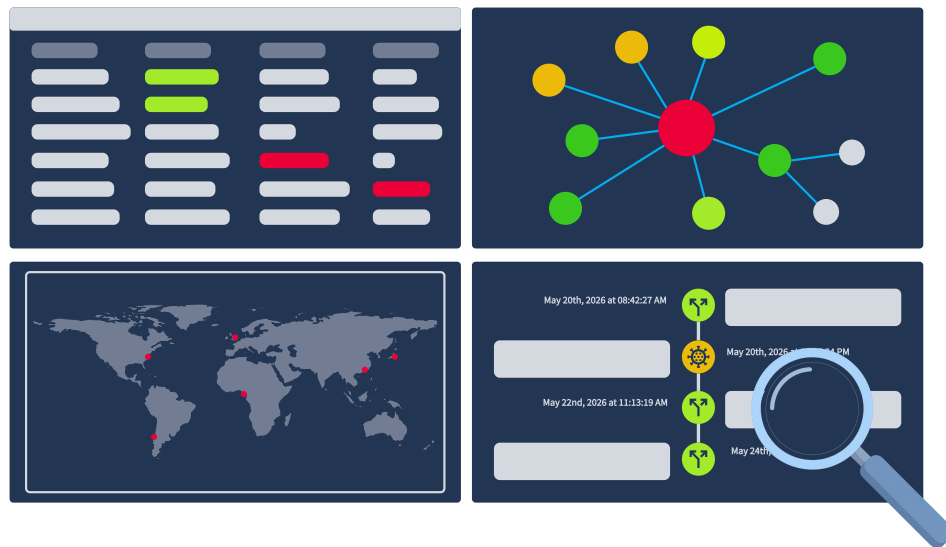
Tool version
-

LATEST CREATED RELATIONSHIPS

Relationship Type	Tool	Target	Date	Score	Action	More
USES	ATTAC...	[T1027.013] Encrypted/Encode...	May 20, 2...	5 - Im...	Copyri...	⋮
USES	ATTAC...	[T1685] Disable or Modify Tools	May 20, 2...	5 - Im...	Copyri...	⋮
USES	ATTAC...	[T1056.001] Keylogging	May 20, 2...	5 - Im...	Copyri...	⋮
USES	ATTAC...	[T1573.002] Asymmetric Crypt...	May 20, 2...	5 - Im...	Copyri...	⋮
USES	INTR...	APT-C-36	May 20, 2...	5 - Im...	Copyri...	⋮

Task 6 - Threat Intel Investigation

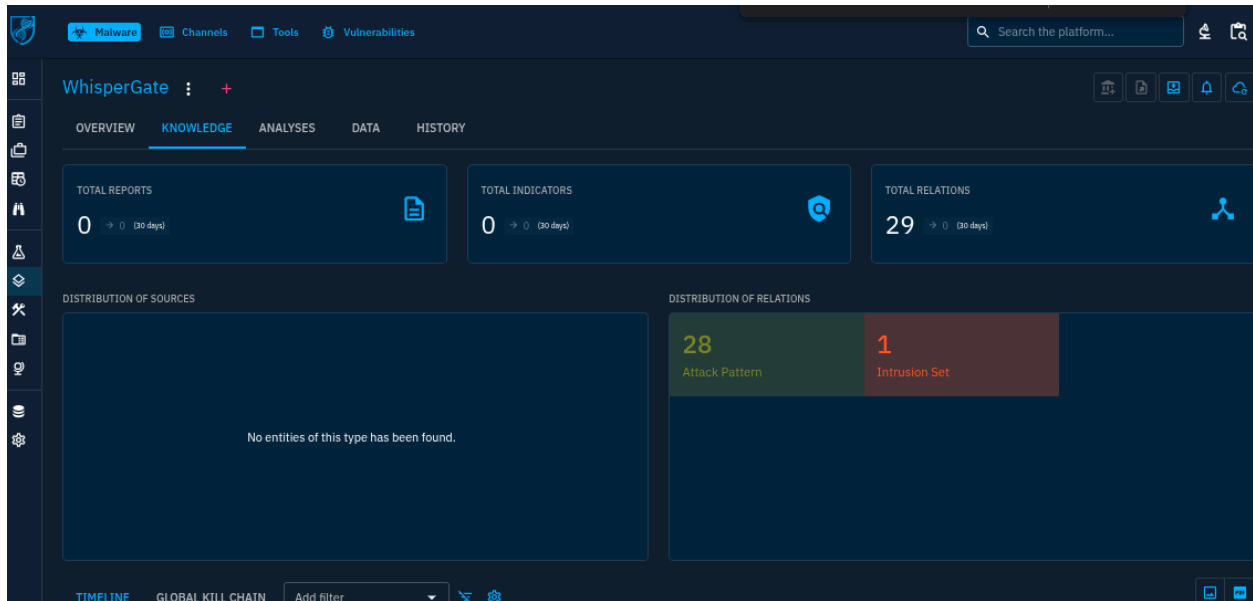
You are a threat intelligence analyst at a global Managed Security Service Provider (MSSP) supporting critical infrastructure clients. A recent wave of destructive attacks has hit your client's related sectors. Your team has been tasked with profiling two emerging threats in your OpenCTI instance: the **WhisperGate** malware family and the **Saint Bear** intrusion set. Investigate and gather data from OpenCTI to build a quick intelligence brief for incident response teams. Focus on timelines and TTPs to support detection efforts. Best of luck!



Answer the questions below

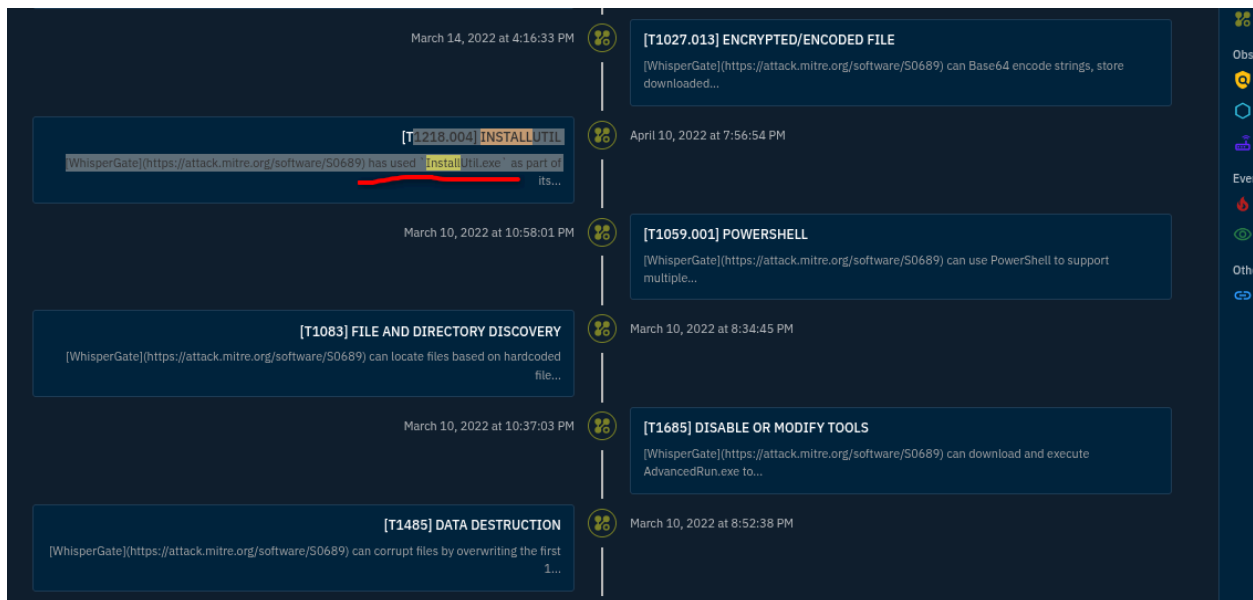
| How many attack pattern relations are linked to the WhisperGate malware?

28



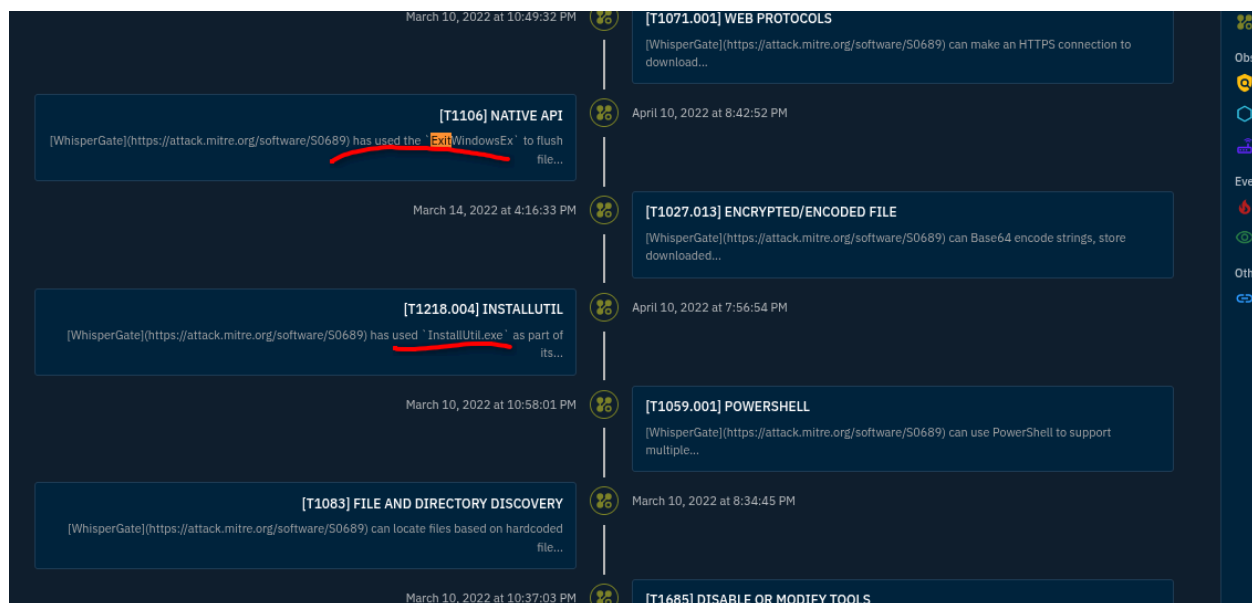
Which Windows utility does WhisperGate use to disable Windows Defender?

InstallUtil.exe



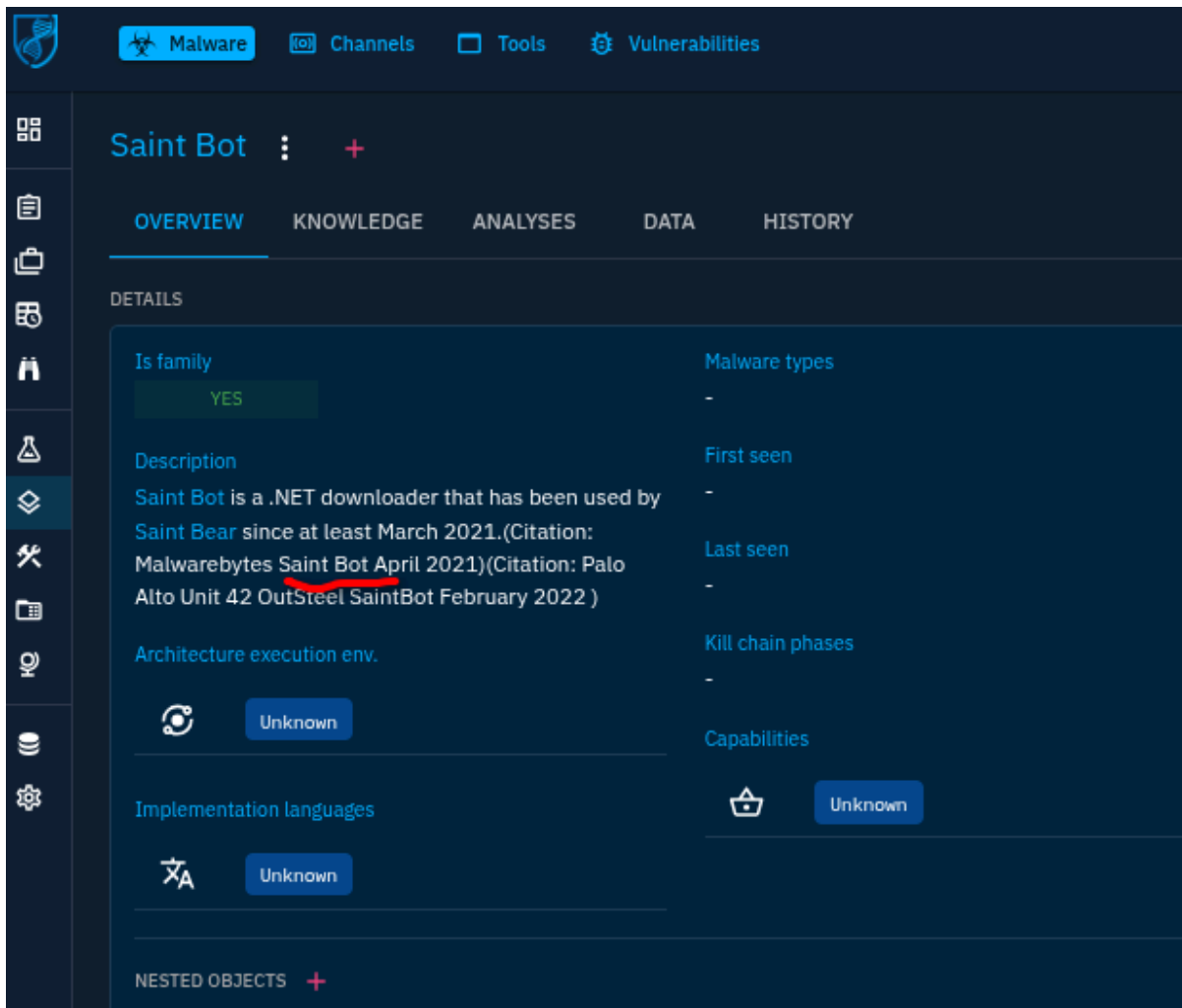
What native API function does WhisperGate use to shutdown a compromised host?

ExitWindowsEx



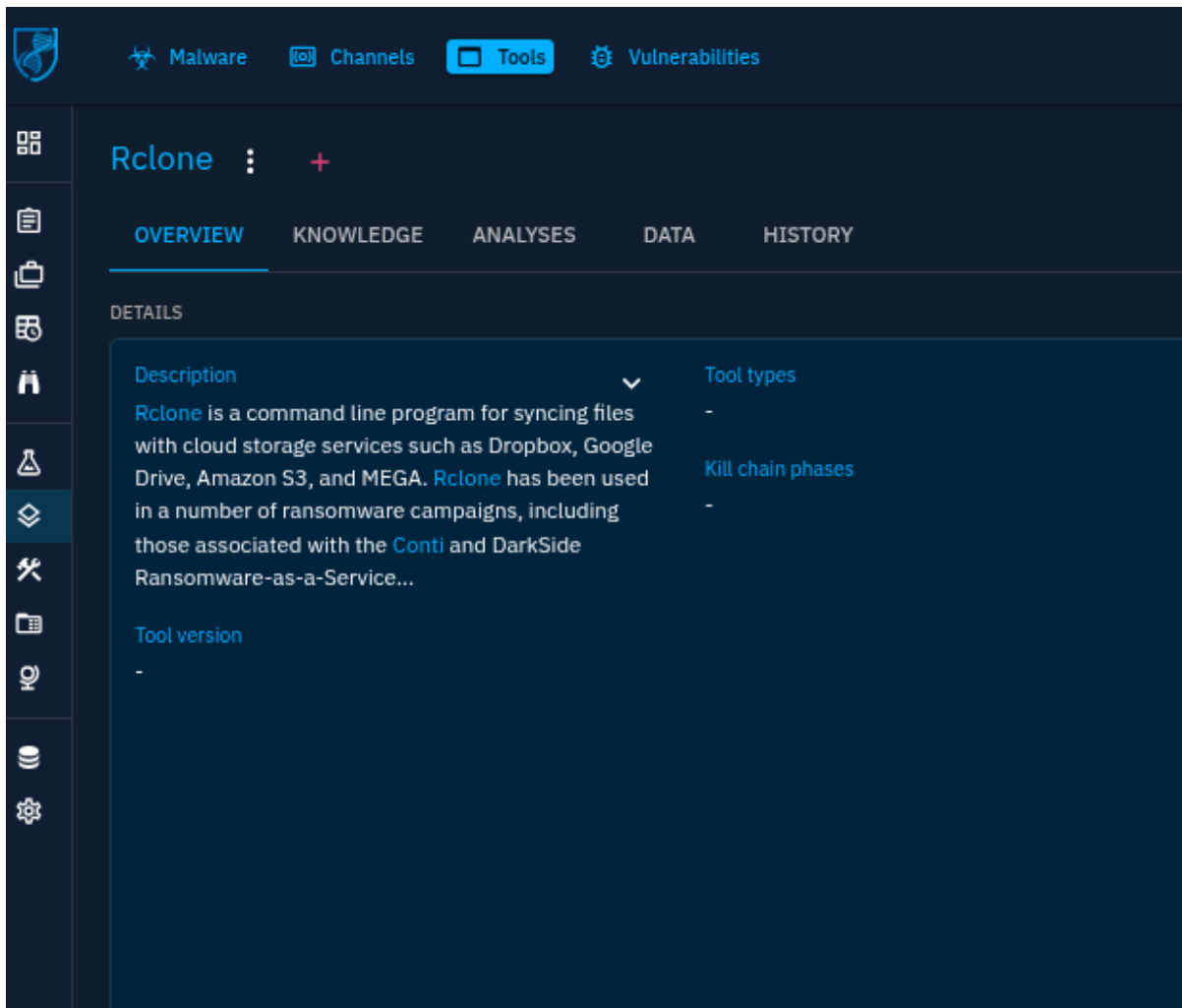
What is the name of the downloader malware commonly paired with OutSteel in Saint Bear campaigns?

Saint Bot



Which tool has Saint Bear used to exfiltrate data to cloud storage services?

RClone



Which Microsoft Exchange vulnerability has Saint Bear been witnessed exploiting?

CVE-2022-41040

Task 7 - Conclusion

In this room, you explored the sources of threat intelligence data and how it is ingested into the platform. You then covered the OpenCTI UI and learned how to efficiently traverse both hot (recent/high-priority) and cold (historical) knowledge sections. Finally, you wrapped it up with a realistic threat intelligence investigation, aiming to understand how specific threat groups operate and the malicious tools they deploy to target organizations.